

**Measuring stigma for key populations:
Protocol for a systematic review measuring stigma affecting sex
workers (SW) and men who have sex with men (MSM)**

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INTRODUCTION AND RATIONALE

Stigma involves deeply discrediting a person or group on the basis of a perceived attribute, behaviour or reputation associated with them [1]. This discrediting creates a gap between one's "virtual" and "actual" social identities [1]. Individuals engaged in sex work (SW) and men who have sex with men (MSM) can face high levels of stigma associated with these identities and behaviours: these key populations are at increased risk for acquiring and transmitting HIV, and stigma can negatively impact additional aspects of their physical and mental health and well-being [2-4]. With growing recognition of the importance of stigma, there has been increased research and programmatic interest and investment in stigma mitigation interventions, particularly in the realm of HIV/AIDS-associated stigma affecting those living with HIV/AIDS [5-9]. Less is known about measures of stigma affecting key populations, such as MSM and SW, whose sexual practices may put them at risk for HIV, although research is being conducted to measure stigma associated with engaging in sex work and/or same-sex sexual practices among men [3,10,11]. Valid and reliable stigma metrics are needed to assess the impacts of interventions and changes in stigma over time. Yet, there remains limited consensus regarding existing and appropriate stigma metrics specific for these key populations, which is what this systematic review seeks to address. Understanding the measurement of SW- and MSM-associated stigma and differences between stigma metrics specific for these key populations can inform and support the evaluation of future research and stigma reduction interventions.

RESEARCH OBJECTIVE AND QUESTIONS

Objective:

To summarize and synthesize studies using quantitative, qualitative or mixed methods to measure stigma associated with individuals engaging in sex work and/or men who have sex with men.

Review Questions:

- 1) How is stigma associated with individuals engaged in sex work and/or men who have sex with men being measured?
- 2) What metrics are being used, and what validated, reliable stigma scales exist for these key populations?

INCLUSION/EXCLUSION CRITERIA

Types of studies

This review will include primary research studies published in or translated to English that use quantitative, qualitative or mixed methods to directly assess stigma associated with engaging in sex work and/or same-sex sexual practices among men. Secondary data analyses will be excluded from this review, as will any non-studies (e.g. opinion pieces, editorials, book chapters, review articles). Studies that do not directly assess stigma or a related concept in either key population of interest will be excluded (e.g. studies which measure an event and attribute its occurrence to stigma without measuring SW- or MSM-related stigma). The review authors will not restrict by primary research studies' designs or methods as long as they are clearly described. Additionally, there will be no restrictions by study duration, setting or country. No study or publication date filters will be applied to the initial search. However, the review team anticipates imposing a date restriction at a

later stage (e.g. publications in the last ten years) to manage the size and scope of this review and to focus on current research and trends in the literature.

Types of study populations and target populations

Study populations: Any study population among whom stigma associated with SW or MSM is being measured will be included, regardless of whether the study population includes the “stigmatizers”, the “stigmatized”, or both.

Target populations: As noted previously in the protocol, this review will include studies that measure stigma associated with at least one of the key populations of interest: any individual reported as engaging in sex work, or males reported as engaging in same-sex sexual practices. Studies that do not directly measure stigma associated with either group will be excluded. Studies with mixed study and/or target populations assessing stigma – for example, examining attitudes towards same-sex sexual practices among men and among women – will not be excluded on this basis, and efforts will be made to extract the data relevant to this review’s key populations of interest.

Definitions of sex work: Although there can be differences in what is conceived to be sex work and whether individuals engaging in such practices identify as sex workers [12,13], this review will use a broad definition for sex work: sexual acts provided in exchange for money or other goods with non-primary partners [13]. No restrictions will be applied to this population regarding biological sex or gender identity, age, race or ethnicity, country or setting, or duration or frequency of engaging in sex work.

Definitions of MSM: In an effort to capture all definitions of men who engaged in same-sex sexual acts with other men, the MSM population concept will be kept deliberately broad and will include: any individuals of biological male sex and/or those with a male

gender identity who report engaging in sexual acts with other men. The authors recognize that within the published literature, some studies may include transgender with cisgender MSM populations [14-16]. As such, the authors will not exclude studies that examined stigma associated with same-sex sexual practices among men by any definition [14-18]: MSM studies that include transgender people will be excluded, however transgender-specific studies will not be included. To be inclusive and acknowledge the heterogeneity between men who have sex with men, this review will include men who have sex with exclusively men, as well as men who have sex with men and women (MSMW) and/or with transgender persons. There will also be no restrictions regarding duration or frequency of engaging in same-sex sexual practices among men.

Types of stigma

Numerous studies have shown that not all individuals engaging in male-male sexual practices or exchanging sex for money identify as being MSM or sex workers [12, 19-21]. Recognizing the heterogeneity among MSM and individuals involved in SW, this review will include studies measuring stigma associated with the *behavioural practices* of engaging in sex work and/or same-sex sexual practices among men – as opposed to focusing exclusively on stigma associated with *identity* as a member of either key population. Stigma is a complex concept that can include different types, domains and levels of stigma. As such, this review will not restrict on these and related aspects of stigma: any potentially relevant study measuring SW- or MSM-associated stigma of any type (e.g. internalized, perceived, enacted, secondary) [6], any domain (e.g. stigmatization drivers, facilitators or manifestations, or intersecting stigma) [8], and at any level (e.g. individual, group, institution, system) will be considered.

Types of outcomes

Primary outcomes: The primary outcome of interest for this review is an understanding of how stigma associated with sex work and stigma associated with same-sex practices among men are being measured, focusing particularly on which metrics are being used.

Secondary outcomes: Additional outcomes of interest include: which metrics are validated and reliable; which stigma scales being used are specific to the target key populations; and which stigma scales have been validated with respect to health outcomes for key populations.

Studies which do not meet contain information pertaining to this review's outcomes of interest will not be included.

SEARCH METHODS FOR IDENTIFICATION OF STUDIES

Electronic searches

The overall search strategy aims to retrieve appropriate quantitative, qualitative, and/or mixed methods studies measuring stigma related to either key population of interest and has already been developed. The primary author completed a survey of existing literature on this topic in order to scope and inform the development of the search strategy and this protocol. The base Boolean search strategy was developed in one database (PubMed), and then adjusted according to each of the other five databases' individual specifications. The search strategy included two concepts: the *key populations of interest* concept (including sex work- and MSM-related terms), and the *stigma* concept. A third concept, *measurement*, was initially considered for the search strategy. However, after testing the search strategy both with and without the third concept, the review authors decided to add in the

measurement lens at the screening stages, as including *measurement* in the initial search resulted in reduced sensitivity, and loss of potentially relevant articles (e.g. studies assessing a stigma-related concept for SW or MSM that did not specify its measurement and/or any metric in the title or abstract).

The search strategy used a combination of free text words and/or title and abstract word searches as well as controlled vocabulary or subject headings appropriate to each database (e.g. MeSH terms in PubMed, descriptor terms in PsycINFO, Emtree terms in EMBASE). Controlled vocabulary terms were examined in each subsequent database for additional synonyms for the population and stigma concepts. The six databases this systematic review will cover include: PsycINFO, PubMed, EMBASE, CINAHL Plus, Global Health, and WHO Global Health Library Regional Indexes (which searches AIM (AFRO), LILACS (AMRO/PAHO), IMEMR (EMRO), IMSEAR (SEARO), and WPRIM (WPRO)). The final search strategy (Appendix 1) was completed in collaboration with the team's informationist, who has methodological expertise in this area, as well as with input from other team members with expertise in epidemiology and the content areas.

Searching other resources

Where feasible, the authors intend that citation lists of included studies will be searched by hand to identify any other appropriate references, such as an earlier key study that may have developed a stigma scale specific to a key population of interest. Forward searching will also be done, looking for potentially relevant articles among papers that referenced the included studies. Currently, there is no planned review of unpublished or non peer-reviewed sources, such as conference abstracts or other grey literature. The authors made this decision based on a combination of the expectation that a comprehensive review of six

major databases would cover a sufficiently broad base, as well as time and resource constraints.

DATA COLLECTION AND ANALYSIS

The final search strategies in Appendix 1 will be run in all six databases at the end of March 2014, with their results uploaded into the EndNote (version X5) citation manager program. De-duplication will occur, keeping references from databases preferentially in the order of: PsycINFO, PubMed, EMBASE, CINAHL Plus, Global Health, and WHO Global Health Library. The review team will then apply a date filter either from 1 January 2000 onwards, or 1 January 2004 onwards. This will be done to balance keeping the scope of this review current (e.g. literature from the past 10-15 years) while maintaining a comprehensive yet appropriately sized database for screening. Additionally, the team is confident that any potentially pivotal articles from prior to the date cut-off will still be identified and captured through back searching the reference lists of included studies.

After de-duplication has occurred and the date filter has been selected and applied, remaining reference types will be checked, with any identified non-article and non-theses entries removed (e.g. book chapters, editorials). All de-duplicated journal articles and theses will then be exported into an Excel spreadsheet for the screening stages.

Selection of studies

Title and abstract review: There will be two title/abstract screenings, as outlined here. In the first title/abstract screening, every title/abstract entry will be independently reviewed by two reviewers – the review’s primary author (AFH), as well as a second author from the team (either MVW or CH). Author pairs will complete a pilot screen of the same 100

title/abstract entries to check inter-reviewer agreement and ensure that there are no issues with the Excel screening system. Periodic spot-checks will be done throughout the process to check consistency of inter-reviewer agreement between each pair (AFH and MVW; AFH and CH). Authors will apply the *measurement* concept in this screening stage, as alluded to earlier in this protocol, and will exclude any entries not meeting eligibility criteria. When in doubt whether a study includes measurement and fully meets the set criteria, studies will be included for a second screening, in an effort to not rule out something potentially relevant. Any discrepancies between reviewers at this stage will be resolved in the following manner: the two authors will discuss discrepancies and try to come to a consensus regarding including or excluding an entry, and if this cannot be achieved, the review team will err on the side of increased sensitivity and include that entry for a second title/abstract screening. For title entries missing abstracts, titles deemed obviously inappropriate will be excluded in the first title/abstract screening (e.g. editorials, off-topic). For remaining entries without an abstract, attempts will be made to retrieve these for screening.

A second title/abstract screening will be done of all studies included after the first screening. The focus of this second screening is on measurement and metrics: as the measurement concept is only being added in the screening stages, a second screen is felt to be advantageous by the primary author (AFH) and the systematic review supervisor (SB). This second round of dual-reviewer abstract screening will ensure studies to be included measure SW- or MSM-associated stigma using a metric, scale, or similar construct, and that studies will be excluded if they do not identify any methods of measurement or do not directly measure stigma (e.g. measuring a purported association with stigma, indirect or

proxy measures of stigma). Discrepancies between reviewer pairs in identifying studies for inclusion and exclusion will be discussed in an attempt to reach consensus. When consensus cannot be reached, the decision whether to include or exclude a particular entry will be made by the reviewers in consultation with the systematic review supervisor (SB). Abstracts passing this second screening will be included for full-text review, with article PDFs retrieved using EndNote's search functionalities where possible, and manual searching for those remaining. Efforts will be made to access all abstracts' full texts, including using alternate search methods or contacting the authors where possible. (See Appendix 2 for title/abstract screening form fields.)

Full text review: The same author pairs (AFH and MVW; AFH and CH) will independently complete full text reviews of studies included after the abstract screenings, to see if they meet eligibility criteria and inclusion for data abstraction. PDFs of articles will be kept in a full-text review library (in EndNote and on hard-drive) and Excel spreadsheets will be used to track studies' inclusion and exclusion. The full-text review stage will follow the same eligibility criteria as outlined previously in the protocol, with the primary focus being to identify studies with identified metrics, scales, or similar constructs used to measure SW- or MSM-related stigma. Studies that do not meet these criteria will be excluded from the review. The same process used to resolve discrepancies after the second abstract screening will be used to resolve any discrepancies between reviewer pairs after the full-text review. (See Appendix 3 for full-text screening form fields.)

For all screening stages, column hiding will be used to avoid reviewer bias; it will not be possible to blind reviewers to study author or journal. A flow diagram outlining the process of study selection and reasons for exclusion will be completed, in accordance with

the PRISMA statement [22]. All studies included after full-text screening will go through data abstraction, and the review authors will attempt to link together any studies with multiple reports, should any exist.

Data abstraction and management

The review team developed a standardized data abstraction form, drawing from recommendations and abstraction forms from a comparable review [8], and review authors with methodological and content expertise in this area. Information for abstraction will relate to one of three general categories: information about the study (e.g. design, characteristics, quality), population (e.g. study population(s), target key population(s)), and stigma (e.g. type, domain, level, metric-related). (See Appendix 4 for detailed listing of items for data abstraction.)

Data will be abstracted from all included studies using the developed standardized abstraction form (Appendix 4), with one author extracting all information and a second, independent reviewer verifying. The reviewers will independently first complete a pilot abstraction on the same one article in each of quantitative, qualitative, and mixed methods types, in order to test the standardized form and check for any inter-reviewer consistency issues with abstraction. Where possible, the reviewers will contact study authors for any clarifications needed or missing information.

Methodological quality assessment of included studies

Methodological quality and risk of bias will be assessed in included studies, with information extracted by one review author and verified independently by another. Drawing from Stangl et al.'s systematic review of HIV-related stigma interventions [8], a modified version of the Downs and Black checklist [23] will be used to evaluate the quality

of randomized and non-randomized quantitative study designs as well as mixed methods studies. Spencer et al.'s quality framework [24] will be used to assess the quality of research evidence in qualitative studies.

The original Downs and Black checklist [23] consists of 27 items that examine five domains: reporting, external validity, bias, confounding and power. Each item is scored as either 0 or 1, with the exception of Power (scored 0 to 5) and one item under Reporting (scored 0 to 2), for a maximum score of 31. However, certain elements of this checklist were deemed not applicable to the majority of study types anticipated for inclusion in this review, given its largely non-clinical nature. As such, these items (#8,14,15,17, and 27) have been removed from the checklist, including the power item (#27) and domain – scored out of 5 – as the review team anticipates few studies will report power calculations. (See Appendix 5 for the modified Downs & Black checklist [23] used.) Thus, the maximum quality score for any study from this modified checklist is 22. The checklist does not have a designated value to identify high- and low-quality studies. This review will use the mid-point of 11 as its cut-off, a strategy employed in a comparable systematic review [8]. Below is an explanation of the four remaining domains in this review's modified Downs and Black [8] checklist (as the power domain is excluded):

- **Reporting:** measures the extent to which information has been reported and allows readers to form an unbiased assessment of a study's results;
- **External validity:** examines the extent that a study's results are generalizable from the study sample to the population;
- **Bias:** assesses biases in intervention and outcome measurements;
- **Confounding:** addresses selection bias in choosing study participants [8].

Spencer et al.'s quality framework [24] will be used to assess the methodological quality of qualitative studies. Their framework employs 18 questions (each scored 0 or 1), divided across the nine subscales outlined below that broadly cover findings, the research process, and general research conduct:

- **Findings:** credibility, extent original research aims have been addressed, clarify of evaluation basis, if findings extends the knowledge base, and scope for wider inference (5 questions);
- **Design:** defensibility of the research design (1 question);
- **Sample:** defensibility of the sample design and composition (2 questions);
- **Data collection:** how well was data collection carried out (1 question);
- **Analysis:** analytic approach described, data source retention and portrayal, diversity of perspective, and analytic detail and complexity (4 questions);
- **Reporting:** clear links between data and interpretations, clarity and coherence in reporting (2 questions);
- **Reflexivity and neutrality:** clarity of assumptions shaping the evaluation (1 question);
- **Ethics:** evidence of attention to ethical issues (1 question);
- **Auditability:** adequacy of documentation of the research process (1 question) [24].

Following from Stangl et al. [8], a cut-off score greater than 9 on the Spencer et al. framework [24] will be used to identify high-quality qualitative studies.

Heterogeneity

The review authors anticipate large methodological, content and statistical heterogeneity across studies, due to the nature and complexity of stigma associated with key populations

and the scope of this review. Methodological heterogeneity is expected given inclusion of quantitative, qualitative and mixed methods designs, which will vary substantially within each category. Large variations in the content of included studies are expected due to differences in study and target key populations, different types, domains and levels of stigma, and variation in chosen metrics to measure stigma. Statistical heterogeneity is anticipated given variability in how the outcomes of interest will be defined and measured across the studies.

Qualitative synthesis

This systematic review will include a flow diagram detailing identification of the included studies at each stage, as mentioned previously in this protocol. General information including sample sizes, location of studies, and the like will be given to give an overview of the types of included studies. This review will then summarize and synthesize the information abstracted, as outlined in the Data abstraction above and in Appendix 4, and present a comparison chart on features of included studies (e.g. a comparison of validated stigma scales). The review will provide a comprehensive account of the included studies, commenting on: study characteristics, design, and methodological quality; characteristics of the study and target populations; and stigma information, including the types, domains, and levels of stigma measured by various stigma scales. The qualitative analysis will discuss how the methodological quality of included studies could have influenced the results of this systematic review.

Given the anticipated heterogeneity, as discussed above, the review team is planning to undertake informed exploratory investigations of the findings, conducting qualitative analyses by subgroups to elucidate trends in measuring SW- and MSM-associated stigma.

Examples of such analyses include examining the data by: types, domains, and/or levels of measured stigma; measuring SW- vs. MSM-associated stigma; study populations; stigma associated with different definitions and types of same-sex practices among men.

Quantitative analysis

A meta-analysis is not warranted for a systematic review of this nature, which lacks standardized definitions, measurement, and reporting of outcomes. However, some basic quantitative investigation can be done, given that measurement of stigma is an outcome of interest. Simple calculations are planned to give a numerical representation of study design, population, and stigma elements (e.g. proportion of all included studies measuring SW-associated stigma).

Unit of analysis issues

The unit of analysis in this systematic review is study level data.

Dealing with missing data

The review team will attempt to contact study authors where possible in the event of any missing or unclear data. Where this is not possible, the review team will attempt to analyze the findings both including the studies in question, and excluding them to see how these studies may impact the results of this review. Imputation of data will be considered where appropriate, with data imputed conservatively so as to not overestimate the results found in this review.

Comments on amendments to the protocol after protocol first developed:

Revision date	Amendment to protocol
May 3, 2014	Removed from Inclusion Criteria, p.5: “Articles which detail this information (e.g., a paper about a proposed study’s methods) but do not yet have published results will still be considered for inclusion and can be analyzed separately later.” Rationale: clarity of wording, and the author team wished to include studies that published results.
May 3, 2014	Removed from Data Collection and Analysis, p. 10: “Depending on the number of potentially relevant titles after de-duplication” Rationale: it was determined at the time of running the search strategies that a date filter would be added on after they were run.
September 5, 2014	In consultation with team members, decision made not to conduct quality appraisals of each individual article for inclusion, given the volume of studies for inclusion, resource constraints, and the scope of the review. Methodological assessment and Modified Blacks List (Appendix 5) left in protocol, but will not be completed for each included study.
September 28, 2016	Typographical error regarding formatting of search strategies was corrected
October 14, 2017	Typographical error was corrected (deletion of duplicate text in Inclusion/Exclusion Criteria, Definitions of MSM section)

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APPENDIX 1: SEARCH STRATEGIES

PsycINFO Search Strategy:

S1	(DE "Homosexuality" OR DE "Male Homosexuality" OR DE "Same Sex Intercourse" OR DE "Transsexualism" OR DE "Transgender" OR DE "Transsexualism" OR DE "Bisexuality" OR DE "Prostitution")
S2	TI ("MSM" OR "MSMW" OR "men who have sex with men" OR "sex with both" OR "male homosexual" OR "male homosexuals" OR "homosexual male" OR "homosexual males" OR "homosexual man" OR "homosexual men" OR "gay men" OR "gay man" OR "gay male" OR "gay males" OR homosexuality OR transgender* OR transsexual* OR "trans man" OR "trans men" OR transman OR transmen OR "trans woman" OR "trans women" OR transwoman OR transwomen OR "trans male" OR "trans males" OR "trans female" OR "trans females" OR bisexual* OR "LGBT" OR "GLBT" OR "sex work" OR "sex worker" OR "sex workers" OR prostitut* OR "transactional sex" OR "sex for money" OR "commercial sex" OR "call girl" OR "call girls") OR AB ("MSM" OR "MSMW" OR "men who have sex with men" OR "sex with both" OR "male homosexual" OR "male homosexuals" OR "homosexual male" OR "homosexual males" OR "homosexual man" OR "homosexual men" OR "gay men" OR "gay man" OR "gay male" OR "gay males" OR homosexuality OR transgender* OR transsexual* OR "trans man" OR "trans men" OR transman OR transmen OR "trans woman" OR "trans women" OR transwoman OR transwomen OR "trans male" OR "trans males" OR "trans female" OR "trans females" OR bisexual* OR "LGBT" OR "GLBT" OR "sex work" OR "sex worker" OR "sex workers" OR prostitut* OR "transactional sex" OR "sex for money" OR "commercial sex" OR "call girl" OR "call girls")
S3	S1 OR S2
S4	(DE "Stigma" OR DE "Shame" OR DE "Prejudice" OR DE "Social Isolation" OR DE "Homosexuality (Attitudes Toward)" OR DE "Social Discrimination" OR DE "Social Perception" OR DE "Social Comparison" OR DE "Stereotyped Attitudes")
S5	TI (shame OR stigma* OR stereotyp* OR prejudic* OR homonegativ* OR homoprejudice OR "social perception" OR "social perceptions" OR discriminat* OR alienation OR homophobi* OR antigay OR "anti-gay" OR "anti-homosexual" OR "anti-homosexuality" OR antihomosexual* OR "social isolation") OR AB (shame OR stigma* OR stereotyp* OR prejudic* OR homonegativ* OR homoprejudice OR "social perception" OR "social perceptions" OR discriminat* OR alienation OR homophobi* OR antigay OR "anti-gay" OR "anti-homosexual" OR "anti-homosexuality" OR antihomosexual* OR "social isolation")
S6	S4 OR S5
S7	S3 AND S6

PubMed Search Strategy:

#1	"Homosexuality, Male"[Mesh] OR "Transsexualism"[Mesh] OR "Transgendered Persons"[Mesh] OR "Bisexuality"[Mesh] OR "Sex Workers"[Mesh] OR "Homosexuality"[Mesh:NoExp]
#2	"MSM"[tw] OR "MSMW"[tw] OR "men who have sex with men"[tw] OR "sex with

	both"[tw] OR male homosexual*[tw] OR homosexual male*[tw] OR homosexual man[tw] OR homosexual men[tw] OR gay men[tw] OR gay man[tw] OR gay male*[tw] OR transgender*[tw] OR transsexual*[tw] OR trans man[tw] OR trans men[tw] OR transman[tw] OR transmen[tw] OR trans woman[tw] OR trans women[tw] OR transwoman[tw] OR transwomen[tw] OR trans male*[tw] OR trans female*[tw] OR bisexual*[tw] OR "LGBT"[tw] OR "GLBT"[tw] OR sex work*[tw] OR prostitut*[tw] OR transactional sex*[tw] OR "sex for money"[tw] OR commercial sex[tw] OR call girl*[tw] OR homosexuality[tiab]
#3	#1 OR #2
#4	"Social Stigma"[Mesh] OR "Prejudice"[Mesh] OR "Shame"[Mesh] OR "Stereotyping"[Mesh] OR "Social perception"[Mesh] OR "Social Alienation"[Mesh] OR Homophobia[Mesh] OR "Social Discrimination"[Mesh]
#5	shame[tiab] OR stigma*[tw] OR stereotyp*[tw] OR prejudic*[tw] OR homonegativ*[tw] OR homoprejudice[tw] OR social perception*[tw] OR discriminat*[tw] OR alienation[tw] OR social isolation[tw] OR homophobi*[tw] OR antigay[tw] OR "anti-gay"[tw] OR anti-homosexual*[tw] OR antihomosexual*[tw]
#6	#4 OR #5
#7	#3 AND #6

EMBASE Search Strategy:

#1	('male homosexuality'/exp OR 'homosexuality'/de OR 'men who have sex with men'/exp OR 'men who have sex with men and women'/exp OR 'transsexuality'/exp OR 'transgenderism'/exp OR 'male to female transsexual'/exp OR 'female to male transsexual'/exp OR 'LGBT people'/exp OR 'bisexuality'/exp OR 'bisexual male'/exp OR 'prostitution'/exp OR 'transactional sex'/exp)
#2	("MSM" OR "MSMW" OR "men who have sex with men" OR "sex with both" OR "male homosexual" OR "male homosexuals" OR "homosexual male" OR "homosexual males" OR "homosexual man" OR "homosexual men" OR "gay men" OR "gay man" OR "gay male" OR "gay males" OR homosexuality OR transgender* OR transsexual* OR "trans man" OR "trans men" OR transman OR transmen OR "trans woman" OR "trans women" OR transwoman OR transwomen OR "trans male" OR "trans males" OR "trans female" OR "trans females" OR bisexual* OR "LGBT" OR "GLBT" OR "sex work" OR "sex worker" OR "sex workers" OR prostitut* OR "transactional sex" OR "sex for money" OR "commercial sex" OR "call girl" OR "call girls");ti,ab
#3	#1 OR #2
#4	('social stigma'/exp OR 'stigma'/exp OR 'prejudice'/exp OR 'shame'/exp OR 'social isolation'/exp OR 'homophobia'/exp OR 'social discrimination'/exp)
#5	(shame OR stigma* OR stereotyp* OR prejudic* OR homonegativ* OR homoprejudice OR social perception* OR discriminat* OR alienation OR homophobi* OR antigay OR "anti-gay" OR (anti NEXT/1 homosexual*) OR antihomosexual* OR "social isolation");ti,ab
#6	#4 OR #5
#7	#3 AND #6

CINAHL Plus Search Strategy:

S1	(MH "Homosexuals, Male") OR (MH "Homosexuals") OR (MH "Homosexuality") OR (MH "Bisexuality") OR (MH "Bisexuals") OR (MH "GLBT Persons") OR (MH "Transsexualism") OR (MH "Transsexuals") OR (MH "Transgendered Persons") OR (MH "Prostitution")
S2	TI ("MSM" OR "MSMW" OR "men who have sex with men" OR "sex with both" OR "male homosexual" OR "male homosexuals" OR "homosexual male" OR "homosexual males" OR "homosexual man" OR "homosexual men" OR "gay men" OR "gay man" OR "gay male" OR "gay males" OR homosexuality OR transgender* OR transsexual* OR "trans man" OR "trans men" OR transman OR transmen OR "trans woman" OR "trans women" OR transwoman OR transwomen OR "trans male" OR "trans males" OR "trans female" OR "trans females" OR bisexual* OR "LGBT" OR "GLBT" OR "sex work" OR "sex worker" OR "sex workers" OR prostitut* OR "transactional sex" OR "sex for money" OR "commercial sex" OR "call girl" OR "call girls") OR AB ("MSM" OR "MSMW" OR "men who have sex with men" OR "sex with both" OR "male homosexual" OR "male homosexuals" OR "homosexual male" OR "homosexual males" OR "homosexual man" OR "homosexual men" OR "gay men" OR "gay man" OR "gay male" OR "gay males" OR homosexuality OR transgender* OR transsexual* OR "trans man" OR "trans men" OR transman OR transmen OR "trans woman" OR "trans women" OR transwoman OR transwomen OR "trans male" OR "trans males" OR "trans female" OR "trans females" OR bisexual* OR "LGBT" OR "GLBT" OR "sex work" OR "sex worker" OR "sex workers" OR prostitut* OR "transactional sex" OR "sex for money" OR "commercial sex" OR "call girl" OR "call girls")
S3	S1 OR S2
S4	(MH "Stigma") OR (MH "Prejudice") OR (MH "Shame") OR (MH "Social Isolation") OR (MH "Homophobia") OR (MH "Social Alienation")
S5	TI (shame OR stigma* OR stereotyp* OR prejudic* OR homonegativ* OR homoprejudice OR "social perception" OR "social perceptions" OR discriminat* OR alienation OR homophobi* OR antigay OR "anti-gay" OR "anti-homosexual" OR "anti-homosexuality" OR antihomosexual* OR "social isolation") OR AB (shame OR stigma* OR stereotyp* OR prejudic* OR homonegativ* OR homoprejudice OR "social perception" OR "social perceptions" OR discriminat* OR alienation OR homophobi* OR antigay OR "anti-gay" OR "anti-homosexual" OR "anti-homosexuality" OR antihomosexual* OR "social isolation")
S6	S4 OR S5
S7	S3 AND S6

Global Health Search Strategy:

1.	(exp homosexuality/ OR exp men who have sex with men/ OR exp bisexuality/ OR exp prostitution/ OR exp prostitutes/ OR exp sex workers/)
2.	(MSM OR MSMW OR men who have sex with men OR sex with both OR male homosexual* OR homosexual male* OR homosexual man OR homosexual men OR

	homosexuality OR gay men OR gay man OR gay male* OR transgender* OR transsexual* OR trans man OR trans men OR transman OR transmen OR trans woman OR trans women OR transwoman OR transwomen OR trans male* OR trans female* OR bisexual* OR LGBT OR GLBT OR sex work* OR prostitut* OR transactional sex* OR "sex for money" OR commercial sex OR call girl*).tw.
3.	1 OR 2
4.	(exp stigma/ or exp social stigma/)
5.	(shame OR stigma* OR stereotyp* OR prejudic* OR homonegativ* OR homoprejudice OR social perception* OR discriminat* OR alienation OR homophobi* OR antigay OR anti-gay OR anti homosexual* OR antihomosexual* OR "social isolation").tw.
6.	4 OR 5
7.	3 AND 6

WHO Global Health Library Search Strategy:

(MSM OR MSMW OR men who have sex with men OR homosexual OR gay OR transgender\$ OR transsexual\$ OR bisexual\$ OR LGBT OR GLBT OR sex work\$ OR prostitute\$ OR transactional sex\$ OR sex for money OR commercial sex OR call girl\$ OR homosexuality) AND (shame OR stigma\$ OR stereotyp\$ OR prejudice\$ OR homonegativ\$ OR homoprejudice OR social perception\$ OR discriminat\$ OR alienation OR social isolation OR homophobi\$ OR antigay OR anti homosexual OR antihomosexual\$)
--

APPENDIX 2: TITLE/ABSTRACT SCREENING FORM FIELDS

Excel spreadsheets for both title/abstract screening stages contain the following information on the entries:

- Year
- Authors
- Title
- Abstract
- Name of Database
- Journal

In separate (hidden) columns, each reviewer completes the following fields, selecting from the options:

- Reviewer name
- Decision (from drop-down menu)
 - Include
 - Exclude
- Reason for exclusion (from drop-down menu)
 - Off topic
 - Not primary research
 - Not primary research-review¹
 - Neither population
 - Not stigma-related
 - Not measuring stigma
 - No measurement methods
 - Other (specify in notes)
 - Duplicate entry (specify # in notes)
 - No metric or scale identified**
- Notes (free text field for each reviewer's notes)
- Agreement between reviewers' decisions (coded using Excel functionalities)
- Final decision

***Note: The "No metric or scale identified" option is only for the second abstract screening*

¹ Note: this option was used as an internal flag for potentially useful review articles for subsequent research endeavours

APPENDIX 3: FULL-TEXT SCREENING FORM FIELDS

Excel spreadsheets for both title/abstract screening stages contain the following information on the entries:

- Year
- Authors
- Title
- Abstract
- Name of Database
- Journal

In separate (hidden) columns, each reviewer completes the following fields, selecting from the options:

- Reviewer name
- Decision (from drop-down menu)
 - Include
 - Exclude
- Reason for exclusion (from drop-down menu)
 - Off topic
 - Not primary research
 - Not primary research-review²
 - Neither population
 - Not stigma-related
 - Not measuring stigma
 - No measurement methods
 - Other (specify in notes)
 - Duplicate entry (specify # in notes)
 - No metric or scale identified
- Notes (free text field for each reviewer's notes)
- Agreement between reviewers' decisions (coded using Excel functionalities)
- Final decision

² Note: this option was used as an internal flag for potentially useful review articles for subsequent research endeavours

APPENDIX 4: STANDARDIZED DATA ABSTRACTION FORM FIELDS

(Note: *italicized* indicate drop-down menu items; underlined indicates free-text information)

General

- Last name of first author
- Title
- Year
- Reviewer initials

Study

Characteristics

- Country(ies) of study
- Language

Methods

- *Quantitative; Qualitative, Mixed Methods*

Primary Study Design

- *Cross-sectional; Cohort (incl. prospective, retrospective); Case-control; Quasi-experimental; Randomized; Randomized-RCT; Randomized-Cluster); Program monitoring/implementation research; Qualitative Design; Other;*

Population

Study population(s)

- Study population includes:
 - *MSM only, MSM+other pop, SW only, SW+other pop, MSM+SW only, MSM+SW+other pop, Other pop only (no MSM or SW), Not specified, Unsure*
- Study population A
 - *Health workers; Students; Teachers; Youth; Community members/leaders; Religious group members/leaders; Family or friends; Stigmatized key pop(s) only*
 - *Other (specify in notes)*
- Describe study samples (incl. n for each group, total N for all groups, where possible)

Focus key population(s)

Key stigmatized populations:

- Target key population(s) stigma is about:
 - *Sex workers only, MSM only, Sex workers and MSM*

Focus key population – individuals engaged in sex work (SW):

- Sex or gender of SWs (select “.” if N/A):
 - *F only SWs, M only SWs, F+M SWs, SWs include transgendered, not specified, .*

APPENDIX 4: STANDARDIZED DATA ABSTRACTION FORM FIELDS (continued)

Focus key populations – men who have sex with men (MSM):

- Definition(s) of the men having sex with other men (select “.” if N/A):
 - *non-transgender MSM only, transgender MSM only, both, not specific.*

Stigma

Stigma Characteristics

- Stigma type A from stigmatized perspective (N/A if stigmatizer perspective)
 - *Internalized, perceived/anticipated, experienced/enacted, courtesy/secondary, other, N/A*
- Stigma type B from stigmatized perspective (N/A if stigmatizer perspective)
Stigma Measures
 - *Internalized, perceived/anticipated, experienced/enacted, courtesy/secondary, other, N/A*
- Stigma type C from stigmatized perspective (N/A if stigmatizer perspective)
 - *Internalized, perceived/anticipated, experienced/enacted, courtesy/secondary, other, N/A*
- Additional stigma(s) intersecting with key pop stigma:
 - *HIV/AIDS, MSM or trans, Sex work, injection drug use, race/ethnicity, multi, other, N/A*
- Stigma primary outcome of interest?
 - *Yes, No*

• Stigma Measures

- Validity (any type) of stigma scale(s)?
 - *Yes, no, partly, ?*
- Reliability of stigma scale(s)?
 - *Yes, yes-Cronbach's alpha given, no, partly, ?*
- Stigma scale 1 (authors, name, year)
- Stigma scale 2 (authors, name, year)
- Other stigma scales used (authors, name, year)

Notes

APPENDIX 5: METHODOLOGICAL QUALITY ASSESSMENT (MODIFIED CHECKLIST)

Modified version of Downs & Black checklist [23].

Note: If an item is N/A for a study, highlight item and that study, score that item N=0:

a) Calculate study's score having all N/A items included with scores of N=0 for each.

b) Calculate study's score having removed all N/A items, and reduce that study's score denominator by the number of N/A items (e.g. if 2 N/A items, -2 from denominator)

Reporting (scored either Yes=1, No=0):

- 1. Is the hypothesis/aim/objective of the study clearly described?
- 2. Are the main outcomes to be measured clearly described in the Introduction or Methods section?
 - If main outcomes first mentioned in Results, score N=0
- 3. Are the characteristics of the participants included in the study clearly described?
 - Cohort studies, trials: inclusion and/or exclusion criteria should be given
 - Case-control studies: case-definition and source for controls should be given
- 4. Are the interventions/exposures of interest clearly described?
- 5. Are the distributions of principal confounders in each group of subjects to be compared clearly described?
 - List of principal confounders is provided
- 6. Are the main findings of the study clearly described?
 - Simple outcome data (including numerators and denominators) reported for all major findings (*Note: this question does not cover statistical tests*)
- 7. Does the study provide estimates of the random variability in the data for the main outcomes?
 - If non-normally distributed data, the inter-quartile range should be reported; in normally distributed data, the standard error, standard deviation, or confidence intervals should be reported.
 - If the distribution of the data is not described, assume appropriate estimates were used and score Y=1.
- 9. Have the characteristics of participants lost to follow-up been described?
 - If no loss to follow-up or losses so small that findings would be unaffected by their inclusion, score Y=1; if no reporting on lost to follow-up, score N=0.
- 10. Have actual probability values been reported (e.g. $p=0.02$ rather than $p<0.05$) for the main outcomes, except when $p<0.001$?

External Validity (scored Yes=1, No=0, Unable to determine=0):

- 11. Were individuals asked to participate in the study representative of the entire population from which they were recruited?
 - Source population and process for patient selection must be identified.
 - Participants are representative if they comprise the entire source population, are an unselected sample of consecutive individuals, or a random sample.

- Studies not reporting the proportion of the source population the participants are derived from should be scored U=0.
- 12. Were individuals who were prepared to participate representative of the entire population from which they were recruited?
 - The proportion of those asked who agreed should be stated.
 - Sample is representative if the study demonstrates that the distribution of the main confounding factors was the same in the study sample and the source population.
- 13. Was the context (places, facilities, exposures) where an intervention/exposure was experienced by participants representative of that of the majority of participants?

Internal Validity - Bias (scored Yes=1, No=0, Unable to determine=0):

- 16. If any results in the study were based on “data dredging”, was this made clear?
 - Any analyses not planned at the outset should be indicated.
 - If no retrospective unplanned subgroup analyses were reported, score Y=1.
- 17. In trials and cohort studies, do the analyses adjust for different lengths of follow-up of patients, or in case-control studies, is the time period between the intervention and outcome the same for cases and controls?
 - If different lengths of follow-up were adjusted for, score Y=1, and if not, score N=0.
 - If follow-up was the same, score Y=1.
- 18. Were the statistical tests used to assess the main outcomes appropriate?
 - If the distribution of the data (normal or not) is not described, assume estimates were appropriate and score Y=1.
 - If little statistical analysis but no evidence of bias, score Y=1.
- 20. Were the main outcome measures used accurate (valid and reliable)?
 - If clearly described and/or refer to other work, score Y=1.

Internal Validity – Confounding (Selection Bias) (scored Yes=1, No=0, Unable to determine=0):

- 21. Were the participants in different intervention groups (trials and cohort studies) or were the cases and controls (case-control studies) recruited from the same population?
- 22. Were participants in different intervention groups (trials and cohort studies) or were the cases and controls (case-control studies) recruited over the same period of time?
- 23. Were participants randomized to groups?
- 24. Was the randomized assignment concealed from both participants and staff until recruitment was complete and irrevocable?
 - All non-randomized studies, score N=0.
 - If assignment not concealed for non-participants, score N=0.

- 25. Was there adequate adjustment for confounding in the analyses from which the main findings were drawn?
 - Score N=0 if: main conclusions based on analyses of treatment (not intention to treat), or if the distribution of known confounders in different groups was not described or was different between groups but not taken into account in analyses.
 - Score N=0 if: non-randomized study that did not investigate and/or did not adjust for confounders.
- 26. Were losses of participants to follow-up taken into account?
 - If too small to affect findings, score Y=1.

Note: Items deemed not applicable or irrelevant that were removed from the original Downs and Black checklist [23] include item numbers: 8, 14, 15, 19, 27 (power item).